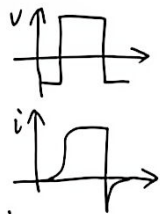
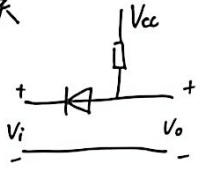
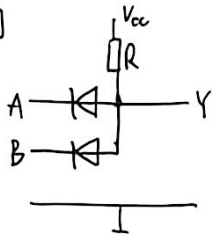


## 二极门电路

开关

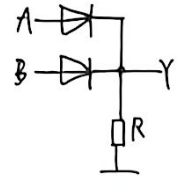


与门



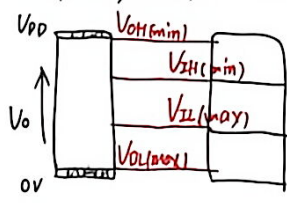
AB一个为0 Y=0

或门



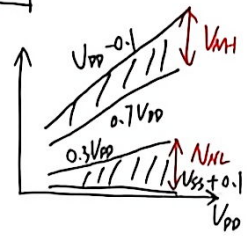
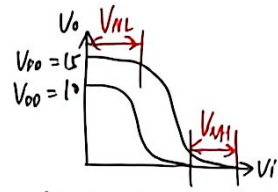
AB一个为1 Y=1

## (2) 输入端噪声容限 (直流)

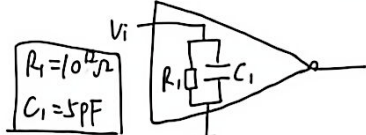
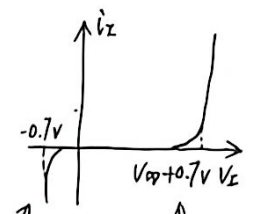
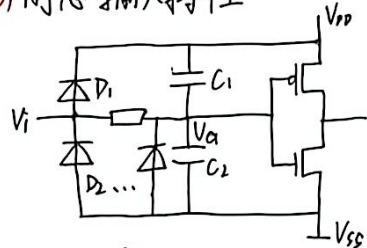


高  $V_{NH} = V_{OH(min)} - V_{IH(min)}$

低  $V_{NL} = V_{IL(max)} - V_{OL(max)}$



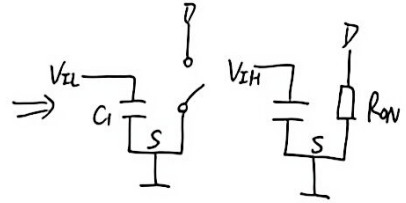
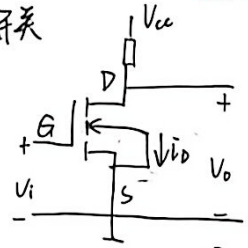
## (3) 静态输入特性



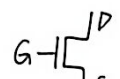
输入电容  $C_i$  上的电压  $< V_{DD} + V_{DF}$

## CMOS 门电路

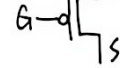
1. 开关



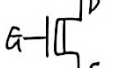
N增强



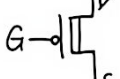
P增强



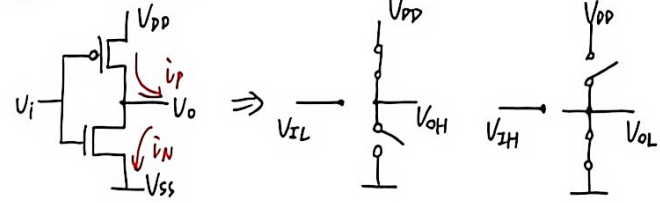
N耗尽



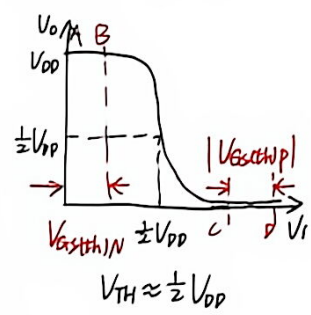
P耗尽



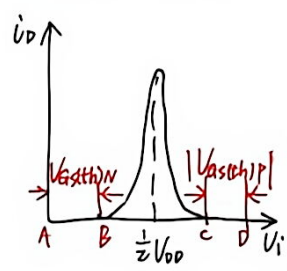
2. CMOS 反相器



## (1) 电压传输特性

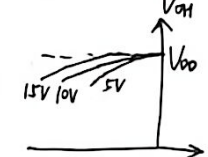
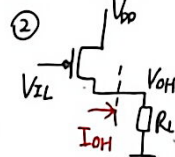
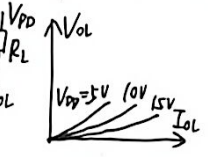
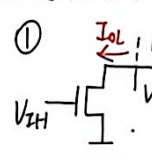


## 电流传输特性



## (4) 静态输出特性

流入芯片的负载电流为正



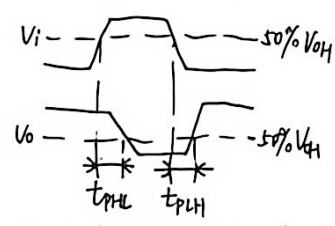
PMOS管漏极特性曲线

$V_{DD}$  上升  $V_{GS}$  上升  $R_{ON}$  下降  $V_{OL}$  下降

$V_{DD}$  上升  $R_{ON}$  下降  $V_{OH}$  下降

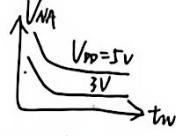
## (5) 动态特性

### ① 传输延迟时间

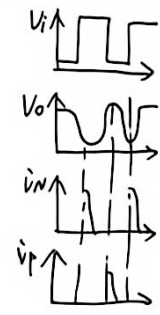


为使  $t_{pd}$  下降  
负载电容下降  
MOS 导通电阻下降  $\Leftarrow V_{TH}$  上升

### ② 交流噪声容限 $\uparrow \Leftarrow t_{pd}$



### ③ 动态功耗

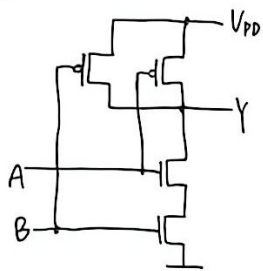


$P_S = I_{DD} V_{DD}$   
 $P_D = P_C + P_T = (C_L + C_{PD}) f V_{DD}^2$   
 $P_{TOT} = P_S + P_D$

功耗电容  
MOS 导通功耗  $P_T = C_{PD} f V_{DD}^2$   
负载电容放电  $P_C = C_L f V_{DD}^2$

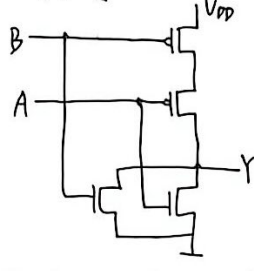
⑭ 输出 数个负载电容 ↑ 开关工作频率 上限 ↓

### 3. 与或非门

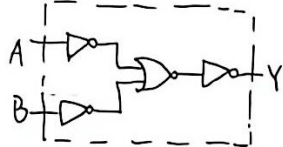
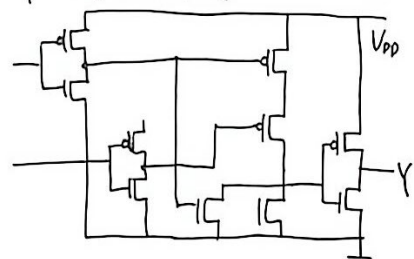


带缓冲级与或非门

### 或非门

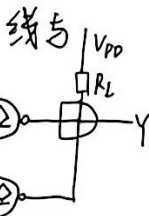
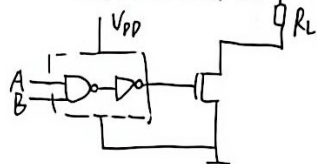


带缓冲级或非门 (类似M)

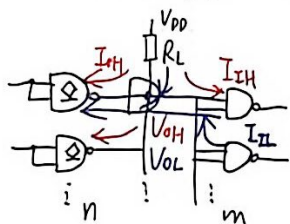


### 4. OD门 (漏极开路)

OD 输出的与或非门

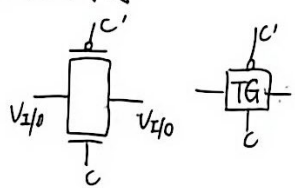


外接上拉电阻范围

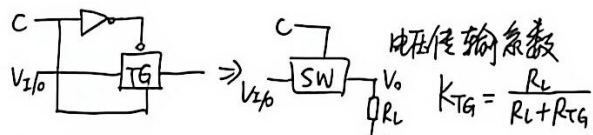
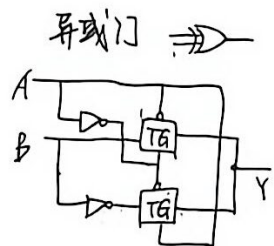


$$\begin{cases} V_{DD} - R_L(NI_{OH} + mI_{IH}) \geq V_{OH} \\ \frac{V_{DD} - V_{OL}}{R_L} + m|I_{IL}| \leq I_{OL(max)} \end{cases}$$

### 5. 传输门



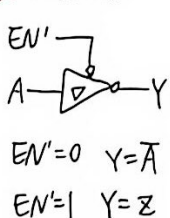
双向模拟开关



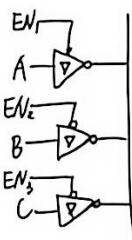
电压传输系数

$$K_{TG} = \frac{R_L}{R_L + R_{TG}}$$

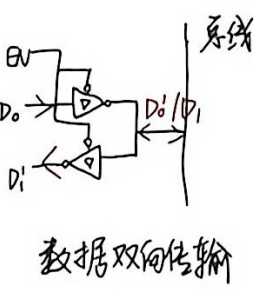
### 6. 三态门



$EN' = 0 \quad Y = \bar{A}$   
 $EN' = 1 \quad Y = Z$



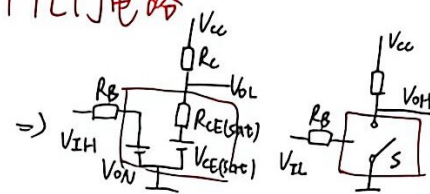
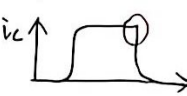
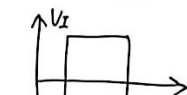
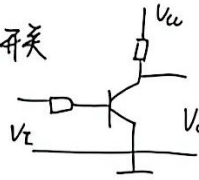
总线



数据双向传输

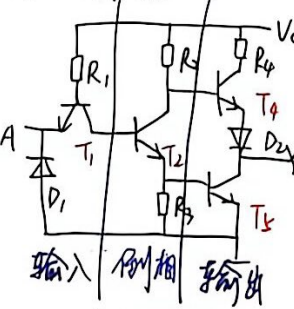
## TTL门电路

### 1. 开关



过渡量经过放大区  
 除了  $V_{OH}$  时  $T_4$  放大区  
 $V_{OL}$  时  $T_1$  倒置放大区  
 其他均在饱和/截止

### 2. TTL反相器

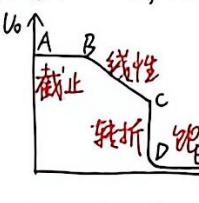


输入 倒相 输出

$V_{OH} = 0.7V$   
 ①  $V_{IL} = 0.2V$   $T_1$  发射结导通  
 $V_{OH}$  钳位在  $0.9V \sim 1.1V$   $T_2 \times$   
 $T_1$  饱和 /  $T_2 \times$  /  $T_4 \checkmark$  /  $T_5 \times$   
 ②  $V_{IH} = 3.4V$  若  $V_{OH} = 4.1V$   
 $T_2$  通  $V_{OH}$  钳位在  $2.1V$   
 $T_1$  倒置 /  $T_2 \checkmark$  /  $T_5 \checkmark$  /  $T_4 \times$

$D_2$ : 确保  $T_5$  饱和导通时  $T_4$  可靠截止  
 $D_1$ : 抑制负干扰脉冲; 防  $V_{IL} < 0$  时发射极电流过大

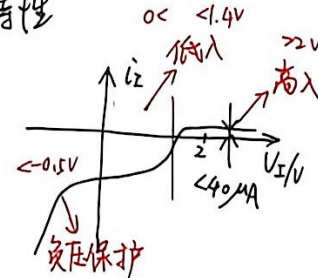
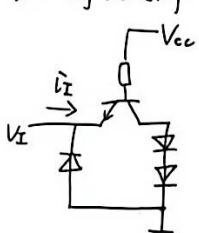
### 3. 电压传输特性



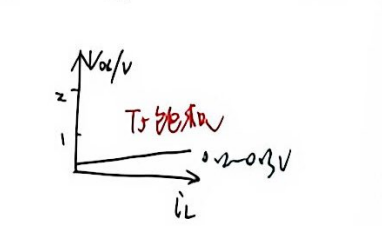
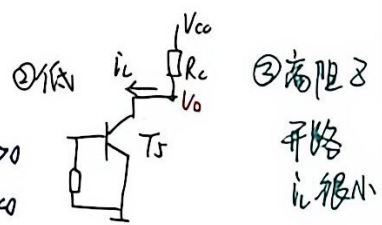
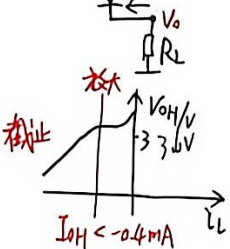
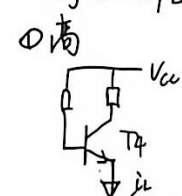
$V_{IL}$   
 $< 0.6V$   $AB: T_1 \times T_2 \times T_5 \times T_4$  饱和  
 $0.7 < 4.3V$   $BC: T_1 \checkmark T_2$  线性  $\checkmark T_5 \times T_4 \checkmark$   
 $> 1.4V$   $CD: T_1 \checkmark T_2 \checkmark T_5 \checkmark T_4 \times$   
 $DE$ : 同上

### 4. 输入噪声容限 (同CMOS)

### 5. 静态输入特性

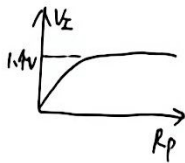
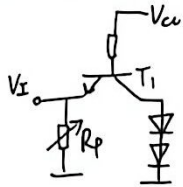


### 静态输出特性



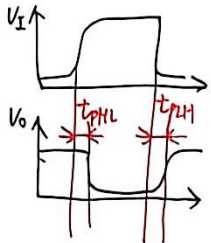


## 6. 输入端负载特性

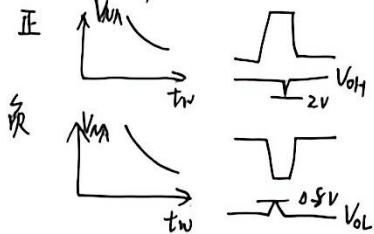


## 7. 动态特性

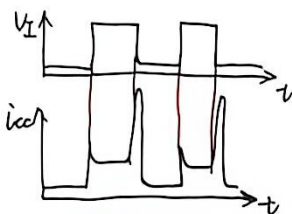
### ① 传输延迟时间



### ② 交流噪声容限



### ③ 动态尖峰电流: T4 T5 同时导通



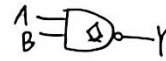
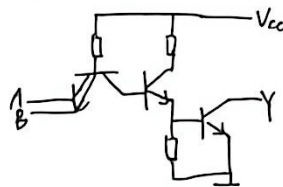
一个周期内尖峰脉冲平均值

$$I_{PMV} = \frac{1}{T} (I_{cm} - I_{ccL}) t_{PH}$$

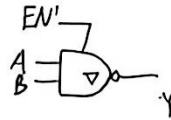
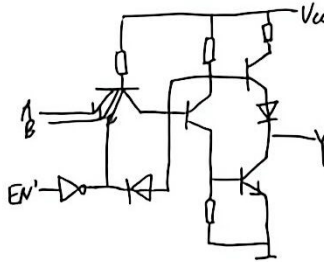
电源电流均值

$$I_{ccAV} = \frac{1}{2} (I_{ccL} + I_{ccH}) + \frac{1}{2} f t_{PH} (I_{cm} - I_{ccL})$$

## 9. OC门



## 10. TS门(三态)



## 逻辑化简

### 运算

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$X \oplus 1 = \overline{X} \quad X \oplus 0 = X$$

$$X \oplus 0 = X \quad X \oplus 1 = \overline{X}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

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$$A \oplus B = \overline{A}B + A\overline{B}$$

$$A \oplus B = \overline{A}B + A\overline{B}$$

### 其它

$$A + AB = A$$

$$A(A+B) = A$$

$$A + A'B = A + B$$

$$AB + A'C + BC = AB + A'C$$

$$AB + A'C + BC = AB + A'C$$

$$AB + AB' = A$$

$$A(AB)' = AB'$$

### 逻辑函数

$$Y = F(A, B, \dots) = \sum m_k = \prod M_k$$

无关项 < 约束项

$$A \oplus B = A \odot B$$

$$\overline{AB} \neq \overline{A} \cdot \overline{B}$$

$$A \odot B = A \odot B$$

$$\overline{A+B} \neq \overline{A} + \overline{B}$$

$$A \odot B = A \odot B$$

只有 A=B 时 < 成立

### 反函数 F

$$\text{对偶: } F' = \left( \begin{matrix} + \leftrightarrow \cdot \\ \cdot \leftrightarrow + \end{matrix} \right)$$

奇数个异或 = 1

偶数个异或 = 0

$$X \odot 0 = \overline{X}$$

### 逻辑运算符

按位运算符 ~ & | ^ ~ 或 ~ ~

反与 或异或 同或

| AB \ CDE | 000 | 001 | 011 | 010 | 110 | 111 | 101 | 100 |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|
| 00       |     |     |     |     |     |     |     |     |
| 01       |     |     |     |     |     |     |     |     |
| 11       |     |     |     |     |     |     |     |     |
| 10       |     |     |     |     |     |     |     |     |